

# ASSESSMENT 6

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**1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.**

## **Architecture Design**

- Use **CDN (Content Delivery Network)** for caching videos near users
- Deploy **multiple application servers** behind a load balancer
- Use **object storage (e.g., cloud storage buckets)** for video content
- Enable **auto-scaling groups** for compute resources

## **Load Balancing & Auto-Scaling**

- Load balancer distributes user requests across servers
- Auto-scaling increases instances during peak traffic
- Ensures no server overload → smooth streaming

## **Caching Techniques**

- CDN caches frequently accessed videos
- Reduces latency and buffering

## **Cost Optimization**

- Use **pay-as-you-go** model
- Auto-scale down during low demand
- Use **spot instances / reserved instances**
- Optimize storage with tiering (hot/cold storage)

**2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and**

**scaling can be handled efficiently.Highlight the benefits and risks of a multi-cloud deployment model.**

### **Proposed Solution**

- Use **Kubernetes** for container orchestration
- Deploy across multiple cloud providers (AWS, Azure, GCP)
- Use **CI/CD pipelines** for consistent deployment

### **Service Discovery & Scaling**

- Kubernetes provides **service discovery (DNS-based)**
- Auto-scaling using **Horizontal Pod Autoscaler (HPA)**
- Load balancing handled internally

### **Benefits**

- Avoid vendor lock-in
- High availability
- Better resilience

### **Risks**

- Complex management
- Security inconsistencies
- Data transfer costs

**3. An e-learning platform experiences data loss due to accidental deletion in the cloud.The organization needs a robust backup and disaster recovery strategy.Design a solution using cloud-native backup, replication, and versioning.Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.Provide steps to ensure business continuity during failures.**

### **Solution Design**

- Use **cloud backup services** with versioning enabled
- Enable **multi-region replication**
- Use automated backup scheduling

## **RTO & RPO**

- **RTO (Recovery Time Objective):** Achieved using fast restore systems
- **RPO (Recovery Point Objective):** Achieved using frequent backups/versioning

## **Business Continuity Steps**

1. Regular automated backups
2. Store data in multiple regions
3. Test recovery process
4. Use failover mechanisms
5. Monitor backup status

**4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.**

## **Architecture Design**

- Sensitive data → On-premises
- Non-sensitive apps → Public cloud
- Secure connection via **VPN or dedicated link**

## **Identity & Security**

- Centralized identity (IAM/SSO)
- Role-based access control
- Data encryption (at rest & in transit)

## **Challenges**

- Complex management
- Integration issues
- Security consistency

- Latency between environments

**5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.**

### **Deployment Strategy**

- Deploy application across **multiple regions**
- Use **global load balancer**

### **Data Replication & Failover**

- Use **database replication (active-active or active-passive)**
- Automatic failover if one region fails

### **Monitoring & Alerting**

- Use cloud monitoring tools
- Set alerts for failures
- Enable auto-healing mechanisms

### **Result**

- Ensures **high availability and fault tolerance**
- System continues even if one region goes down